

PHYSICAL MODELING



UNIT 5

Varsha R Jenni

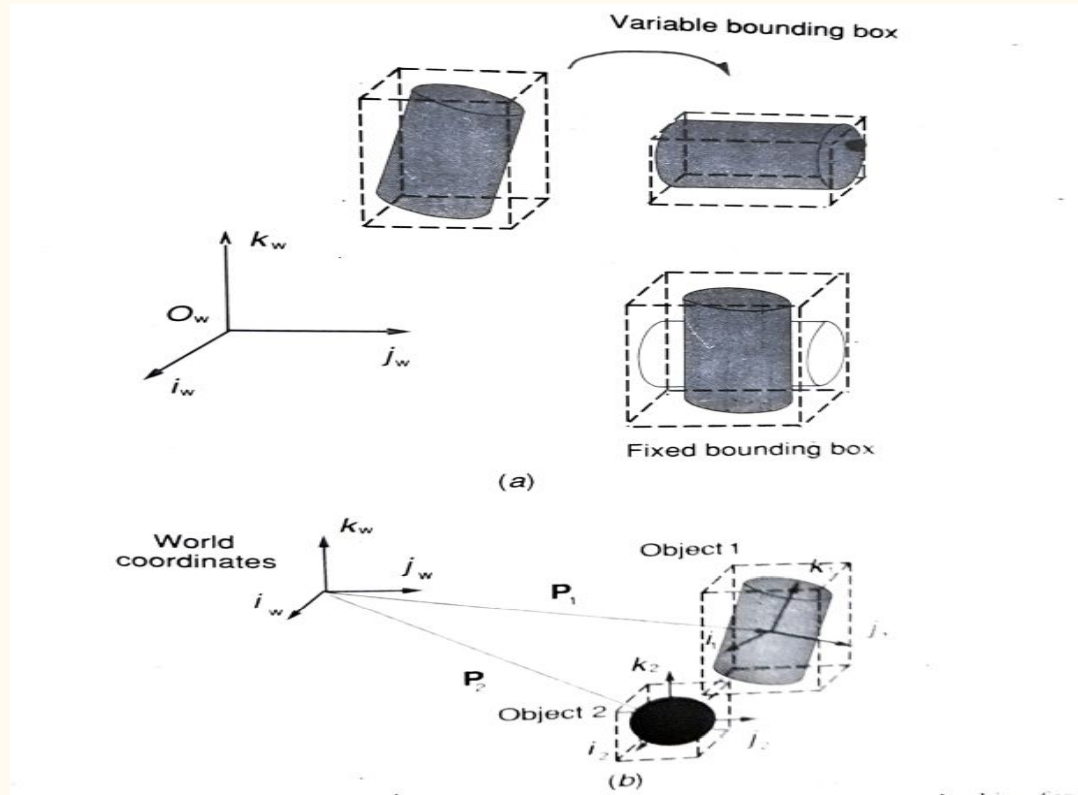
INTRODUCTION

- Physical Modeling - Integration of object's physical characteristics
 - Includes weight, inertia, surface roughness, compliance (hard or soft), deformation mode (elastic or These plastic), etc
- Physical modeling brings in more realism to the virtual world model.
- The computation load required by physical modeling is assigned to the haptics rendering pipeline.
- The stages of haptics rendering pipeline are:
 - Collision detection,
 - Force calculation,
 - Force smoothening
 - Force mapping
 - Force haptic texturing

COLLISION DETECTION

- The first stage of haptic rendering is collision detection
 - determines whether two (or more) objects are in contact with each other
- Collision detection can be classified into
 - Approximate(bounding box collision detection)
 - Uses 3D extents (bounding boxes)
 - A bounding box is a prism which encloses all the vertices of a given 3D object
 - Bounding boxes are classified into
 - oriented
 - An oriented bounding is a prism aligned with the object's major axes and oriented which changes orientation dynamically as the object rotates.
 - axis-aligned
 - Exact

COLLISION DETECTION



FORCE COMPUTATION

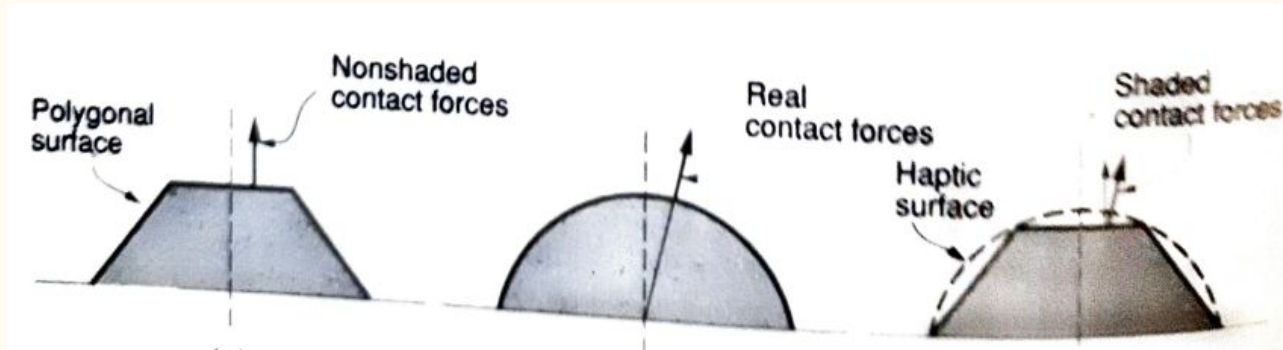
- When users interact with 3D object surfaces (whether these are deformable or not) they should feel reaction forces.
- These forces need to be computed by the haptic rendering pipeline and sent to the haptic display providing force feedback to the user
- Force computation takes into account
 - the type of surface contact
 - Single-point contact (eg: PHANToM)
 - Multiple Contact (eg: Haptic Glove)
 - the kind of surface deformation,
 - Rigid bodies that are immobile (eg: walls)
 - Elastic objects that are mobile (eg: rubber ball)
 - Plastically Deformed Objects (eg: coda cans)
 - as well as the object's physical and kinematic characteristics.

FORCE SMOOTHING AND MAPPING

- If we assume that the object is frictionless and that a single point of interaction exists, then direction of the resistive force is that of the surface normal at the point of contact.
- This simplified approach creates unwanted force discontinuities for curved polygon. surfaces. This is due to the abrupt change in surface normal direction from one polygon to the next.
- Therefore the object will feel tessellated, or facetted, when rendered haptically, even if its graphics image looks smooth.

FORCE SMOOTHING AND MAPPING

- **Force shading** changes the direction of the feedback force produced during interactions with polygonal surfaces in order to simulate contact with smooth curved surfaces.
- The direction of the contact force is computed based on the weighted normals of the vertices of the polygon being contacted.
 - does not affect the magnitude of the contact force, only its direction



FORCE SMOOTHING AND MAPPING

- The force magnitude is given by Hooke's law:

$$F_{\text{smoothed},i} = \begin{cases} K_{\text{object}} \cdot d \cdot \overline{N} & \text{if } 0 \leq d \leq d_{\text{max}} \\ F_{\text{max}} \cdot \overline{N} & \text{if } d_{\text{max}} < d \end{cases}$$

- This equation assumes a single point of contact, which works for stylus-based interface interfaces, such as the PHANToM.
- The current location of the point is called the haptic interface point (HIP)
- The efficiency of HIP-based force shading grows with the increase in surface polygon count and decrease in surface curvature.

FORCE SMOOTHING AND MAPPING

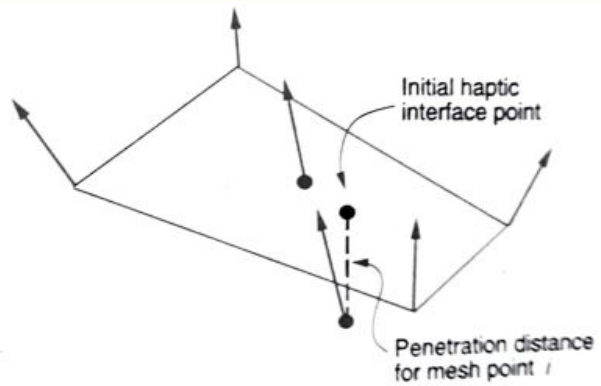
- A haptic glove is usually mapped to a virtual hand, thus the geometry of the virtual fingertip needs to be taken into account.
- In the haptic domain this corresponds to a cylindrical haptic mesh centered on the fingertip.
- Assuming the glove has not reached its saturation point, haptic mesh point the contact force is given by

$$F_{\text{haptic mesh } i} = K_{\text{ball}} \cdot d_{\text{mesh } i} \cdot \overline{N}_{\text{surface}}$$

- $d_{\text{mesh } i}$ is the penetrating distance at mesh point i .
 - This distance between the initial contact point (called "ideal haptic interface point") and the current location of the mesh point;
 - N_{surface} surface normal of the contact polygon at the contact point.



(a)



(b)



FORCE SMOOTHING AND MAPPING

Commercial force feedback gloves, such as the CyberGirasp, have a single actuator per finger.

Therefore the haptic mesh forces need to be summed and the resultat mapped mapped to the corresponding actuator, as in

$$F_{\text{displayed}} = \frac{\sum F_{\text{haptic mesh}}}{\cos(\theta)}$$

where θ is the angle between the haptic glove actuator and the resultant mesh fore

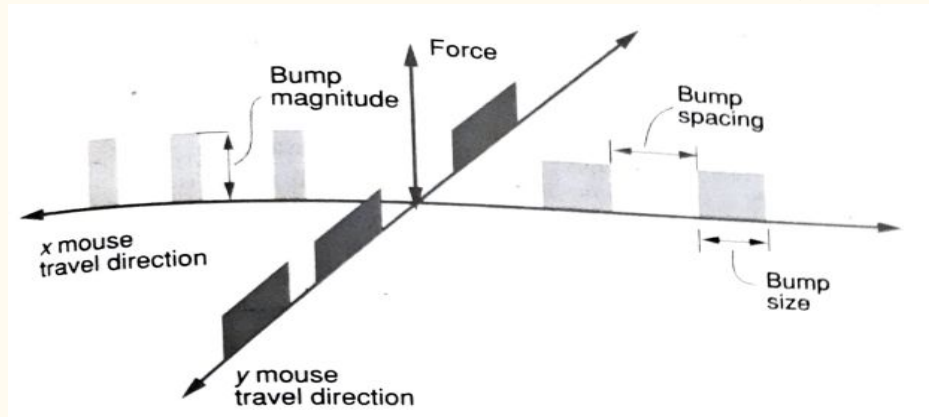
HAPTIC TEXTURING

- Add realism to the object appearance, haptic textures enhance the realism of the physical model of the object surface.
- Haptic textures can add new information to characterize an object as slippery, cold, smooth etc.
- Haptic textures can be cascaded creating new surface effects, similar to the multitexturing approach described previously for graphics.

HAPTIC TEXTURING

Haptic Textures Produced by Tactile Mice

- These interfaces can apply small forces in z direction (normal to the desk surface).
- As a consequence, the surfaces that can be simulated are frictionless.
 - Since no resistive forces can be applied in the horizontal (x - y) plane



HAPTIC TEXTURING

Haptic Textures Produced by Tactile Mice

- The iFeel mouse can produce texture patterns of the type of the height, width and spacing of the bumps.
- The height determines the magnitude of the vertical force mouse actuator, up to the saturation limit.
- The spacing and width of the haptic ridges determine the texture frequency component felt when the mouse pointer scans the object surface.
- Textures can be programmed to change depending upon whether the mouse moves in the positive or the negative direction

HAPTIC TEXTURING

Haptic Textures Produced by the PHANToM

- These textures have a richer content, since this interface can produce forces in any direction.
- As a consequence, friction forces can be added to contact forces in order to simulate sticky surfaces.
- Viscosity can be simulated by multiplying the surface velocity with a damper while inertia can be modeled by a simple mass-acceleration product.
- Surface texture can be created with displacement maps, similar to the bump maps used in multitexturing.
- The displacement map models the cross section of the surface through hills and valleys and the force felt by the user is affected by the local gradient of the surface

HAPTIC TEXTURING

Haptic Textures Produced by the PHANToM

$$F_{\text{texture } x} = A \frac{\Delta h}{\Delta x}$$

$$F_{\text{texture } y} = A \frac{\Delta h}{\Delta y}$$

Where x and y are the translation directions, A is a proportionality constant, and h is the surface height above the surface plane.

Thus $\Delta h / \Delta x$ and $\Delta h / \Delta y$ are the local gradients in the x and y directions respectively.

HAPTIC TEXTURING

Haptic Textures Produced by the PHANToM

- The PHANToM can superimpose high-frequency vibrations on the force feedback signal felt at the user's fingertip.

$$F_{\text{texture}} = A \cdot \sin(mx) \cdot \sin(ny)$$

- A is the constant giving the magnitude (amplitude) of vibrations.
- m and n are constants that modulate the frequency of vibrations in the x and y directions respectively: and x and y are in the range of $(0, 2\pi)$
 - Higher is the value of m and n, higher is the frequency felt by the user
- Overall force felt by user:

$$F_{\text{resist}} = F_{\text{constraint}} + F_{\text{texture}} + F_{\text{friction}}$$

Thank you